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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 03/12/2005 |
| **PI Name:** | Nordtvedt, Kenneth L. Ph.D. | | |
| **Project Title:** | Optimizing Science From a STEP Mission | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FUNDAMENTAL PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Gravitational & relativistic physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | knordtvedt@bresnan.net | **Fax:** | FY 406-522-7656 |
| **PI Organization Type:** | INDUSTRY | **Phone:** | 406-522-7656 |
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| **Zip Code:** | 59715 | **Congressional District:** | 1 |
| **Comments:** | Email changed from kennordtvedt@imt.net (3/07) | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 00-HEDS-02 |
| **Start Date:** | 05/09/2001 | **End Date:** | 05/09/2005 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JPL |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | 0 | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | None | | |
| **Performance Goal No.:** | 1.24 | | |
| **Performance Goal Text:** | To explore the frontiers of physics and uncover mysteries of the universe obscured by Earth's gravity | | |
| **Task Description:** | Study STEP signal fine structure due to oblateness of earth and earth surface irregularities; and relate this Equivalence Principle violating fine structure signal to theory Add additional scientific information about fundamental physical law which can be achievable from a STEP mission. This mission tests one of the fundamental properties of physical law, and additional information from fine structure will aid theorists in interpreting any breakdown of the Equivalence Principle | | |
| **Rationale for HRP Directed Research:** | | | |
| **Research Impact/Earth Benefits:** | | | |

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| **Task Progress:** | My work has revealed additional information and clues that could be learned about the structure of physical law by performing a test of the equivalence principle (universality of gravitational free fall) in Earth orbit. Violation of this principle first put forth by Einstein would also lead to this experiment measuring new details about the Earth's interior. If America were to perform this experiment, it would put her once again at one of the frontier's in fundamental physics exploration. My recent work on comparing gravitational free fall for different isotopes of the same material could be applied to both the space experiment or the American laboratory experiments of the equivalence principle. |
| **Bibliography Type:** | Description: (Last Updated: 12/14/2020) |
| **Abstracts for Journals and Proceedings** | Turyshev S, Shao M, Williams JG, Nordtvedt K, Murphy T. "The Lunar Astrometric Test of Relativity (LATOR) Mission." 2004 NASA/JPL Workshop on Physics for Planetary Exploration, Solvang CA, April 2004.<br>2004 NASA/JPL Workshop on Physics for Planetary Exploration. , Apr-2004 |